1 (a) Simplify $m^3 \times m^4$

Add the powers: 3 + 4 = 7, so $m^3 \times m^4 = m^7$.

1 mark
Add powers for the same base

m^7

(1)

(b) Simplify $(5np^3)^3$

$(5np^3)^3 = 5^3 \times n^3 \times p^9 = 125n^3p^9$

2 marks total
Cube every factor; multiply powers of p

$125n^3p^9$

(2)

(c) Simplify $\dfrac{32q^9r^4}{4q^3r}$

32 ÷ 4 = 8; 9 − 3 = 6; 4 − 1 = 3.

2 marks total
Divide coefficients and subtract powers

$8q^6r^3$

(2)

(Total for Question 1 is 5 marks)